OPERATION Department
SIMHADRI

Note No :1

Ref:SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Date: 19/05/2024(11:13:21)

Subject:LMI ON PROCEDURES FOR TESTING AND MAINTENANCE TO AVOID EXCESSIVE OVERSPEEDING OF PRIME MOVERS

Subject: REVISION: 6 OF LMI ON PROCEDURES FOR TESTING TO AVOID EXCESSIVE OVER SPEEDING OF PRIME MOVERS

LMI APPROVAL DOCUMENT

Operation LMI (**LMI/OD/OPS/SYST/003)** on “PROCEDURES FOR TESTING AND MAINTENANCE

TO AVOID EXCESSIVE OVERSPEEDING OF PRIME MOVERS” has been reviewed and revised as per revised OD

“PROCEDURES FOR TESTING & MAINTENANCE TO AVOID EXCESSIVE OVERSPEEDING OF PRIME MOVERS

“OD/OPS/SYST/003 Rev no-02”.

LMI and respective OD is attached and put up for approval.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION) ,Simhadri (RANASINGHBALGIR@NTPC.CO.IN, 9425219715)
On:19/05/2024(11:13:21) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:2

As discussed, for review and necessary correction please.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION) (MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri
On:23/05/2024(16:29:32) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:3

To be modified as per Attached Format.

Submitted by:

सतीश कुमार मुरपाक - वरिष्ठ प्रबंधक

SATISH KUMAR MURAPAKA - 101773 (SENIOR MANAGER ,OPERATION)
(SKMURAPAKA@NTPC.CO.IN, 9437479788) ,Simhadri

On:04/06/2024(13:13:31) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:4

Dear Sir,

LMI modified as desired as per new format. May kindly check.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION)
(RANASINGHBALGIR@NTPC.CO.IN, 9425219715) ,Simhadri

On:22/06/2024(16:44:14) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:5

forwaded for approval.

Submitted by:

सतीश कुमार मुरपाक - वरिष्ठ प्रबंधक

SATISH KUMAR MURAPAKA - 101773 (SENIOR MANAGER ,OPERATION) (SKMURAPAKA@NTPC.CO.IN, 9437479788) ,Simhadri
On:23/06/2024(17:20:11) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:6

For review and necessary approval please.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION) (MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri
On:27/06/2024(10:57:03) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:7

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:27/06/2024(16:43:23) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:8

Dear Sir,

LMI modified as desired as per new format. May kindly review and necessary approval please.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION)
(RANASINGHBALGIR@NTPC.CO.IN, 9425219715) ,Simhadri
On:29/08/2024(05:18:40) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:9

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:29/08/2024(09:25:05) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:10

Unauthorized Use Prohibited

Dear Sir,

LMI modified as per new format. May kindly review and necessary approval please.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION)
(RANASINGHBALGIR@NTPC.CO.IN, 9425219715) ,Simhadri
On:31/08/2024(12:38:53) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:11

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:31/08/2024(17:02:14) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:12

Dear Sir,

LMI has been modified as per new format. May kindly check and put up for approval.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION)
(RANASINGHBALGIR@NTPC.CO.IN, 9425219715) ,Simhadri
On:13/10/2024(21:13:00) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:13

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:15/10/2024(18:28:38) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:14

Dear Sir,

LMI has been modified as per New Format. May kindly check and put up for approval.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION)
(RANASINGHBALGIR@NTPC.CO.IN, 9425219715) ,Simhadri
On:09/11/2024(03:52:57) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:15

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri
On:11/11/2024(09:40:02) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:16

LMI has been reviewed, forwarded for approval.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG) (NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:14/11/2024(18:25:23) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:17

Checked and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION) (SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri
On:20/11/2024(19:26:57) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Unauthorized Use Prohibited

Note No:18

Please review the LMI

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE) (NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri
On:21/11/2024(16:00:15) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:19

LMI has been reviewed, forwarded for approval.

Submitted by:

के पी रमेश - वरिष्ठ प्रबंधक

KANADE PIYUSH RAMESH - 101897 (SENIOR MANAGER ,MECH MAINT- TURBINE) (PIYUSHKANADE@NTPC.CO.IN, 9425219272) ,Simhadri
On:23/12/2024(16:36:18) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:20

Forwarded for approval

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE) (NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri
On:23/12/2024(19:34:40) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:21

May pl review the LMI

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT) (JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri
On:26/12/2024(18:38:09) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:22

Submitted by:

सुनेता साहा - उप महाप्रबंधक

Suneta Saha - 006843 (DY. GENERAL MANAGER ,C & I MAINT) (SUNEITASAHA@NTPC.CO.IN, 9434737713) ,Simhadri
On:26/12/2024(19:24:16) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:23

LMI has been reviewed, forwarded for approval.

Submitted by:

चन्दन कुमार निराला - उप महाप्रबंधक

CHANDAN KUMAR NIRALA - 102031 (DY. GENERAL MANAGER ,C & I MAINT)
(CHANDANKUMARNIRALA@NTPC.CO.IN, 9425178488) ,Simhadri
On:28/12/2024(18:53:28) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:24

Submitted by:

सुनेता साहा - उप महाप्रबंधक

Suneta Saha - 006843 (DY. GENERAL MANAGER ,C & I MAINT)
(SUNEITASAHA@NTPC.CO.IN, 9434737713) ,Simhadri
On:30/12/2024(16:37:06) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:25

Forwarded for further approval

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri
On:01/01/2025(21:06:58) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:26

Reviewed. Forwarded for approval.

Submitted by:

भास्कर वी के कुरीचेट्टी - उप महाप्रबंधक

Bhaskar V K Kurichety - 008045 (DY. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri
On:02/01/2025(15:44:47) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:27

Dear Sir,

LMI reviewed and Forwarded for approval after changing approval route.

Submitted by:

राणा सिंह बलगीर - वरिष्ठ प्रबंधक

RANA SINGH BALGIR - 101855 (SENIOR MANAGER ,OPERATION)
(RANASINGHBALGIR@NTPC.CO.IN, 9425219715) ,Simhadri
On:08/02/2025(11:26:19) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:28

Forwarded for approval.

Submitted by:

भास्कर वी के कुरीचेट्टी - उप महाप्रबंधक

Bhaskar V K Kurichety - 008045 (DY. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri
On:08/02/2025(11:29:28) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:29

Reviewed and submitted for approval please.

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MAINTENANCE)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri
On:08/02/2025(15:26:50) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Unauthorized Use Prohibited

Note No:30

LMI is in line with the OD. May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri
On:08/02/2025(15:38:05) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:31

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri
On:09/02/2025(13:35:52) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

Note No:32

Approved & Submitted by:

समीर शर्मा - मुख्य महाप्रबंधक

Sameer Sharma - 005041 (CGM ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri
On:10/02/2025(19:49:06) **Ref:**SMTPP/OPERATION/2024-25/LMI_OVSP/143709

NTPC SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Procedures for Testing to Avoid Excessive Over Speeding of Prime Movers

LMI/OD/OPS/SYST/003 Rev:6

NOVEMBER 2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	**LMI NAME**	PROCEDURES FOR TESTING TO AVOID EXCESSIVE OVER SPEEDING OF PRIME MOVERS
2	**LMI No.**	LMI/OD/OPS/SYST/003 Rev:6 NOVEMBER 2024
3	**Reference Document**	COS-ISO-00-OD/SYST/003/Rev:3 SEPTEMBER:2024
4	**Superseded Document**	LMI/OD/OPS/SYST/003 Rev:5 MAY 2022
5	**Prepared By**	Name: RANA SINGH BALGIR Senior Manager (Operation)
6	**Checked By**	Name: MALAY KANTI JANA AGM (Operation)
7	**Checked By**	Name: SOMNATH BHATTACHARYA AGM (Operation)
8	**Checked By**	Name: NAVIN KISHORE AGM (EEMG)
9	**Checked By**	Name: NILANJAN MANDAL AGM (TMD)
10	**Checked By**	Name: JAYAMOL RACHEL VARUGHESE AGM (C&I)
11	**Checked By**	Name: BHASKAR V K KURICHETY DGM (MTP)
12	**Rechecked & Reviewed By**	Name: OM PRAKASH AGM (Maintenance)
13	**Rechecked & Reviewed By**	Name: PRASANTA KUMAR JENA GM (O&M)
14	**Approved By**	Name: SAMEER SHARMA CGM (Simhadri)

INDEX

TITLE	Procedures for Testing & Maintenance to Avoid Excessive Over Speeding of Prime Movers
LMI No.	LMI/OD/OPS/SYST/003/REV:6

1.0 Introduction

Over speeding results in higher and undesirable stresses in the rotating assemblies resulting higher residual life consumption and may lead to catastrophic failure in the event of excessive speed rise. So over speeding of prime mover shall be restricted to the absolute minimum in magnitude, duration and frequency to ensure safety of the prime movers and in no case, speed shall rise beyond acceptable limit. Prime movers are likely to get damaged by excessive over speeding, in the event of failure of turbine to trip, arising from

I. The governor failing to arrest the speed rise after load is rejected,
II. Failure of the overspeed tripping devices,
III. Mechanisms Sticking,
IV. The release of stored energy to the prime mover.

The risk of turbine-Generator unwanted over speeding can be mitigated by employing protection trip systems that act rapidly to isolate the steam supply in the event of separation of prime mover from the Grid.

2.0 Superseded Document

Superseded: LMI/OD/OPS/SYST/003 REV:5 MAY 2022

3.0 Details of Revision and Reference Document

Procedure and Periodicity of overspeed testing of prime mover (Main Turbine and TDBFP Drive turbine) provided with Mechanical as well as Electrical / Electronic overspeed system has been made more specific and objective.
Clause no. 6.0/8.0 Preconditions for Main Turbine/ TDBFP drive modified.
New Checklist Sl no.11.0: Prepared and Attached.

This LMI is based on operation directive Procedures for Testing to Avoid Excessive Over Speeding of Prime Movers COS-ISO-00-OD/SYST/003/Rev:3 September:2024

4.0 Scope

This is applicable to 4X500 MW Steam Turbines and TDBFP drive Turbines installed at NTPC Simhadri plant.

5.0 Instructions for Mechanical / Hydraulic System of Main Turbine

Over speeding should be restricted to the absolute minimum magnitude, duration and frequency necessary to ensure safety of the Prime mover. The failure of prime movers following excessive over speeding results from the increased centrifugal forces. However, any over speeding results in higher, and therefore undesirable stresses in the rotating assemblies.

Definition of Overspeed: An over speed is a speed greater than the maximum rotational speed value that the prime mover can achieve when under normal Governor Control. At NTPC Simhadri, the Turbines are provided with an online-testing device for checking the healthiness of over speed protection device.

To prevent over speeding of 500 MW turbines, the healthiness of protective devices is to be tested and ensured during:

I. Normal operation
II. After an overhaul or long shutdown
III. While shutting down the unit.

5.1 During Normal Operation

The following tests shall be done during normal operation of the unit:

Sl. No.	Test	Method	Frequency of testing	Responsibility	Records
5.1.1	Exercising Turbine stop and control valves	To be tested using ATT program. Or Through Starting device–off load condition. (MS/HRH)lines should be in depressurized condition.	After every overhaul: before rolling After shutdown > 1month before rolling	Commissioning &HOD(TMD) HOD(Oprn) &HOD(TMD)	SCE log book Protocol to be made.
5.1.2	Over speed trip device oil injection test	ATT Or Local-from turbine governing rack.	Once in a month/ opportunity. After every overhaul: Before synchronization. After shutdown > 1month: Before synchronization	Shift Charge Engineer Commissioning & HOD(TMD) HOD(Oprn) & HOD(TMD)	SCE log book Protocol to be made.

5.1.3	Remote trip device	Through ATT program	Once in a month or opportunity After every overhaul: Before synchronization After shutdown > 1 month: Before synchronization	Shift Charge Engineer. Commissioning & HOD(TMD). HOD(Oprn) &HOD(TMD)	SCE log book Protocol to be made.

5.1.1 Procedure for Test of Exercising Main Turbine Stop and Control Valves

When the turbine is operated continuously, the stop valves, and frequently the control valves too, remain at the same lift for prolonged periods. Impurities from the steam and the ambient atmosphere are deposited on the valve spindles. They carbonize into hard deposits, which can seriously affect the free movement of the spindles. All stop and control valves must therefore be exercised regularly and cleaned of deposits at the same time.

Testing Requirements:

Exercising can be performed with the automatic turbine tester while the turbine is shut down condition by simulating the start conditions.
OR
With the turbine shut down, exercising is affected by opening and closing the stop valves via the starting and load limiting device, or after the valves have been opened, a trip is initiated to effect closure.

5.1.1.1 Through ATT:

I. Each stop valve is tested together with its associated control valve through ATT.
II. Only one valve assembly may be selected and tested at a time.
III. There must not be any pressure in the supply steam lines when exercising is carried out in this manner.
IV. Select the valve group to be tested.
V. Start the ATT SGC program for valves testing.
VI. First the control valve closes.
VII. When the control valve is completely closed, the stop valves start closing.
VIII. Once stop valve is completely closed, the reverse process starts.
IX. After the valves return to their normal positions, carry out testing for other valves.

5.1.1.2 Through Starting Device:

I. There must not be any pressure in the supply steam lines when exercising is carried out in this manner.
II. Close secondary oil lines to Control valves, CRH NRV and other extractions NRV
III. Reset the Turbine with the help of C&I
IV. Keep test valve of stop valves in open position.
V. Bring down starting device to zero position to build trip oil pressure
VI. Then Operate the starting and load limiting device till~42%
VII. Observe opening of all stop valves.
VIII. Simulate any Turbine trip condition to check closing of all stop valves
IX. Now Close test valve of all Stop valves, & sec. oil to CRH NRV and other extractions NRV
X. Reset the Turbine with the help of C&I.
XI. Bring down starting device to zero position to build trip oil pressure
XII. Then Operate the starting and load limiting device till all control valves open
XIII. Simulate any Turbine trip condition to check closing of all control valves
XIV. Normalize Test valve of Stop valves, secondary oil lines to CRH NRV and other extractions NRV.

5.1.2 Procedure for Test of Overspeed Trip Device Through Oil Injection Test:

The healthiness of this device can be tested by either of the following method:

I. From control room by operating the Automatic Turbine Tester.
II. From local Main Turbine governing rack.

5.1.2.1 From Control Room Through ATT:

I. Select the overspeed devices and start the program in ATT
II. Record the turbine speed, Aux trip fluid pressure and test oil pressure (local) at which the strikers actuate.
III. Confirm from local that both devices have acted.
IV. Confirm the alarm in alarm page when the trip acts.
V. Confirm that both devices got reset & aux trip fluid recovered to its previous value.
VI. Ensure that there is no drop-in aux trip fluid pressure after resetting.
VII. The observations are to be recorded in Protection Check Register.

5.1.2.2 From Local Turbine Governing Rack:

I. Record readings of Aux. trip fluid, test oil pressure, turbine speed, HP & IP secondary oil pressure before performing the test.
II. Release testing device for operation & check for reset oil pressure (>8 ksc) by pressing reset valve (KA02) (right side).
III. Isolate aux trip oil by keeping test valve (KA03) (left side) pressed down & should not be released till the test is complete, else it will lead to draining of aux trip fluid & turbine will trip.
IV. Pressurize test oil circuit by turning test signal transmitter hand wheel (KA01) (center) in clockwise direction gradually.
V. Note the turbine speed and test oil pressure at which the strikers operate. Confirm alarm in control room.
VI. Depressurize test oil circuit by turning test signal transmitter hand wheel (KA01) (center) in Anticlockwise direction.
VII. Reset the overspeed trip device to operating position by pressing the reset valve (KA02) (right side). Verify aux. trip fluid pressure is as before start of the test. Keep reset valve pressed at least for 1 min after restoration of aux trip fluid pressure. Slowly release reset valve.
VIII. Gradually release the test valve (KA03) (left side)
IX. Check all oil pressure have recovered to previous values.
X. Record turbine speed, and test oil pressure at which the strikers have acted in Protection check register.

Note:

I. If the oil injection test results are not satisfactory, the cause shall be investigated immediately and if not resolved, the turbine shall be taken out of service immediately and actual overspeed test done.
II. If actual overspeed test is successful, then until oil injection test gear is rectified, actual overspeed test shall be done every 3 months.

Responsibility: HOD (Opn) / Commissioning/ HOD(C&I) / HOD (TMD)

III. Under base load or part load conditions, frequency correction or/and frequency influence are "on", the load will vary many times during the day depending on grid frequency thus ensuring healthiness of speed governor and control valves. Turbine shall always be rolled through EHC, such that the governor shall always be active to control the speed from the lowest speed.

Responsibility: UCE

IV. Healthiness of hydraulic controller shall be checked every month by taking it into service for adequate duration every month and recorded in UCE logbook.

Responsibility: UCE

5.1.3 Remote Trip Device Checking

I. Start the SGC program for safety devices checking
II. Select the RTS1/RTS2 and give ON command.
III. Upon ON command, Pretest is conducted to check the healthiness of the test oil circuit.
IV. Once Pretest is found healthy, Main test will start and main trip device gets isolated/blocked to prevent actual tripping.
V. Respective RTS is energized and aux trip oil & main trip oil gets drained from main trip device.
VI. Subsequently, after two seconds, RTS is de-energized and reset solenoid gets energized to reset main trip device and further buildup of trip oil & aux trip oil pressure.

5.2 After an Overhaul or Long Shutdown:

The following are to be carried out after an overhaul or long shutdown (> 1 month):

Sl.No.	Test	Method	Precautions	Responsibility	Records
5.2.1	To establish smooth operation of stop, control and governor valves before rolling.	Governing characteristics to be checked over the full range of operation of the valves.	Care to be taken to ensure steam is not admitted into the turbine.	HOD (Opn)& HOD(TMD)	Protocol to be made
5.2.2	Steam tightness test of turbine control valves.	Reset turbine and open stop valves. Observe turbine speed. Turbine speed should not increase by more than 30 rpm.	This is to be done only after reaching rolling parameters and TG on barring gear.	HOD (Opn)& HOD(TMD)	Protocol to be made.

5.2.3	Overspeed trip test by oil injection.	To be done through ATT program or from local.	To be done at 3000rpm before synchronizing the machine.	HOD (Opn)& HOD(TMD)	Protocol to be made.
5.2.4	Healthiness of remote trip device.	After resetting the turbine, trip from control room. Ensure both devices have tripped (both channels).	To be done with main steam line in depressurized condition.	HOD (Opn) & HOD(C&I)	Protocol to be made. (This is a part of turbine protection check)
5.2.5	Correctnes s of turbine speed indicators	Shaft speed should be measured with tachometer/strob oscope and compared with speed indicator readings in control room.	Shaft speed to be measured and compared at 3000rpm. This is to be done before synchronization.	HOD (MTP) &HOD(C&I)	Measured and indicated speeds to be recorded in communicat ion Register.

5.3 While Shutting Down the Unit

The following tests shall be done while unit is being shut down as per indicated frequency of testing:

Sl. No.	Test	Method	Frequency of testing	Responsibility	Records
5.3.1	Steam tightnes s of Turbine valves	Whenever turbine trips before generator (i.e. other than A1 & A2 protection) confirm tripping of generator on low forward power protection within 4 sec.	As and when unit trips.	HOD (Opn)& HOD(TMD)	Trip analysis

5.3.2	Actual Overspeed trip.	Raising turbine speed to permissible overspeed and conforming tripping of the turbine.	As Simhadri Turbines are fitted with both mechanica l and electronic over speed protection system.	HOD (Opn) /HOD(TMD)	Protocol to be made.
1. If any servicing /repair / replacement is carried out in emergency governor or Gap between striker & lever adjusted during TG Overhaul, then actual over speed test mandatorily has to be conducted in next available Opportunity. 2. Electronic simulated overspeed test to be carried out during opportunity shut down once in a year after conductance of actual over speed test till next servicing/repair/adjustment of emergency governor as mentioned.					

5.3.1 Steam Tightness of Turbine Valves:

During normal operations of the turbine and particularly after extended periods of running, the cones and seats of the control valves and stop valves can become damaged by foreign objects or by erosion. The result is that the controllability of the turbine in the range below or at the rated speed is adversely affected, i.e. the uncontrolled admission of steam can give rise to undesirable increases in speed.

During start-up, this situation can give rise to the speed remaining longer than necessary in the critical range, because it is impossible to lower the speed of the turbine. When the passing through the valves is exceptionally severe, it can cause difficulties in synchronizing because the turbine speed is greater than the system speed.

Slight passing at the control valves which causes the speed of the turbine to increase when the stop valves are opened is a normal feature of the valves (balancing holes in the valve cone and clearance at the cone guides) and is therefore not dangerous.

Passing at the valves must be detected early and rectified as soon as possible in order not to endanger the turbine or to adversely affect its availability. Consequently, it is necessary to test the passing of the hydraulically- operated main steam valves at regular intervals.

Testing Requirements:
The passing of the control valves and stop valves can be tested during start-up, during trip of the Turbine or when shutting down the turbine by ensuring Generator trip on low forward power protection.

Test Procedure:

Test Of Control Valves Passing:

I. After reaching rated parameters for Turbine rolling, reset the Turbine
II. Open the Stop valves.
III. Observe the speed increase and record the reading together with the main steam conditions.

Test Of Stop Valves Passing:

I. After reaching rated parameters for Turbine rolling, reset the Turbine.
II. Close the test valves of stop valves.
III. Open the Control valves
IV. Observe the speed increase and record the reading together with the main steam conditions.

5.3.2 Over Speed Trip Procedure While Shutting Down Unit:

Preconditions:

I. Turbine should be warm enough to have achieved more than 80 % of full load expansions.
II. Healthiness of overspeed trip device has been confirmed by oil injection testing within 15 days and more than 1 day prior to carrying out actual over speed test.
III. Two independent speed indicators shall be available for the test. One the speed indicator (Control room & turbine governing rack) and the other a digital tachometer at local.
IV. Generator circuit breaker and field breaker shall be open.
V. Turbine controls shall be continuously manned during the test. The person manning the turbine shall trip the turbine if the speed exceeds 3350rpm.

5.3.2.1 Test Procedure-1: (If Unit to be De-Synchronized)

Prechecks:

I. Ensure C&I/TMD/MTP/EMD personnel are present before start of test.
II. Oil injection test to be done at least 1 day before De-synchronization
III. Ensure turbine vibrations, expansions and bearing temperatures are normal. Record the readings in Shutdown log register
IV. Ensure all the gauges at local are working & values are same as MMI readings.

Deviation if any may be recorded.

V. Ensure turbine speed channels, local value & stroboscope values are within comparable limits.

VI. HP-BP fast opening from GCB/generator breaker open & Turbine trip to be kept bypassed. And HPBP pressure set point to be kept in auto.

VII. Electrical over speed trip set point to be raised after desynchronizing the unit.

VIII. Ensure availability of Mills B, C, D, E and oil guns AB, CD, EF.

Procedure:

I. Start load reduction as per schedule.

II. All persons at desk should be aware of the over speed test procedure.

III. Take unit controller out of CMC at 275 MW & Reduce the unit load to 200MW with B, C, D, E Mills.

IV. Keep MDBFP in service & unload both TDBFPs.

V. Reduce boiler firing and load gradually. Keep two mills in service preferably Mill-B and Mill C. Take oil support in AB & CD elevation, bring burner tilt to 50% for better flame stability.

VI. Charge HP/LP bypass at 180 MW and Keep machine in Load controller mode.

VII. Charge unit bus from station bus at 100MW (for Stage-II only).

VIII. Firing & HP-LP BP should be adjusted in such a way that drum pressure is maintained at 55 -60 ksc. MS/HRH temperature are to be maintained normal (537 deg C).

IX. SH /RH temp to be controlled in such a way that SH temp do not fall to tripping limits.

X. Reduce load to 50 MW by HP/LP bypass.

XI. FD 16 (high range drum level control valve) can be closed before desynchronization as drum level control will be difficult with the same at low loads.

XII. Desynchronize the machine by opening generator breaker/GCB manually and watch for speed rise (if any) and record. Keep field breaker closed till the M/C speed stabilizes at 3000 rpm then open field breaker manually**. (Never bypass turbine trip from generator trip)**

XIII. Ensure that electric over speed trip value raised to 112%, 3360 rpm and Normalize after the test is over. **Responsibility: UCE /HOD(C&I)**

XIV. Watch HPT exhaust steam temperature & maintain re-heater pressure accordingly.

XV. Turbine controller to be taken in hydraulic mode and EHC is to be isolated for over speed test.

XVI. **EHC Isolation Procedure**:

   i. Speeder gear should be at 100%

   ii. Switch off tracking device and raise starting device to maximum (100%).

   iii. Slowly increase the speed reference to maximum (speed reference blocked at 3200 rpm) and record the maximum speed achieved through EHC.

iv. Reduce the turbine speed slowly (not below 2850 RPM as DN/DT can operate) by reducing speeder gear from MMI/ governing rack, observe speed drop w.r.t speed reference /dip in aux sec oil pressure and gradual increase in EHC demand.

v. If speed drops with lowering of speeder gear it indicates that m/c is in hydraulic. Observe if EHC saturates at 100%, then isolate EHC (both valves).

NOTE: If actual speed and speed reference difference is more than 36 rpm then speed reference is blocked and EHC demand can saturate to around 60-70%. After ensuring machine is in hydraulic mode, EHC can be isolated.

XVII. After ensuring turbine controller in hydraulic mode, Raise SG to 100 % slowly, with this, the speed should raise up to 107% of rated speed i.e. 3210 RPM.

XVIII. Further speed raise will be done through the Trip Test Lever very slowly provided on the left side of the speeder gear (Moving towards the EHC isolation valve side). Observe the speed at digital meter (local) and UCB.

XIX. Once the turbine speed reaches to around 3300rpm it should trip on the mechanical over speed protection.

XX. Check both the trip devices are operated.

XXI. If the Machine does not trip at desired speed (110 %) then hand trip the Machine at 112 % speed (3360 RPM).

Activities After the Test is Completed:

I. Normalize the Elect OS Trip protection which was bypassed.

II. Normalize the EHC isolation valves in the Gov. rack.

III. Observe the coasting down of the Machine speed.

IV. Observe other turbo-supervisory Parameters.

V. Record primary oil pressure & TG speed at which trip operated in record book.

Responsibility: UCE

VI. Put Electrical over speed set point at normal value

Responsibility: UCE / HOD(C&I)

VII. All protocols are to be made in Protection Interlock test Register

Responsibility: SCE

5.3.2.2 Test Procedure-2: (If Unit Is Not on The Bar)

After rolling the turbine to 3000 rpm, previous procedure to be followed after EHC isolation. However, all the pre-checks to be followed.

6.0 Instructions for Electrical/Electronic System for Main Turbine

Preconditions:

Electronic over speed test of main turbine shall be done once in a year at turbine speed range 3000 RPM(+/-3%)in turbine warm conditions while taking care of expansions. Machine should be warmed up sufficiently so that casing expansion are at least 80% of full load expansion /or as per OEM recommendation. Expansion readings are to be taken and logged. This avoids over speeding in cold condition and stress on cold rotor.

Procedure For Electronic Overspeed:

Actual Over speed electrical Test is to be carried out by actually raising the speed of the turbine to 3000RPM (+/- 3%) and by lowering the Over speed protection limit set point from 3360 RPM to 3000 RPM (+/- 3%). Note down the speed at which the electric over speed trip signal is generated. Later, reset the electrical trip valve to its original trip value (3360 rpm).

7.0 Actual Over Speed Testing of TDBFP Drive Turbines

Actual Over Speed Testing shall be carried out once in every 12 Months.

Preconditions:
BFP turbine actual over speed test is carried out during overhauling after stopping of the unit. Actual over speed test is carried out if there is no overhauling activity planned on that particular BFP turbine. This test shall be done only if during operation of TDBFP, no abnormal behavior of drive turbine like high bearing vibrations, abnormal bearing temperatures etc. are noticed.

Prerequisite For TDBFP Solo Run:

I. After stopping the BFP ensure it comes to barring. Then isolate the seal steam and wait till the casing metal temperature comes to less than 120 deg C.
II. After that stop barring gear and permit to be issued to TMD to remove the coupling between the BFP turbine and the gear box.
III. Remove the coupling sleeve in between the BFP turbine and the main pump after it has been disengaged. (Coupling removal will take 2 hrs by

TMD). **Responsibility: HOD(TMD)**

IV. Put sticker on turbine shaft for speed measurement with stroboscope.
Responsibility: HOD(MTP)

V. After the couplings have been removed and TMD clearance put BFP turbine on barring by slightly opening the turning gear valve manually and so that the speed is at around 350 rpm. If the turning gear valve is fully opened, then speed will be very high near about 800 rpm.

Procedure:

Following steps to be followed if TDBFP solo run is planned after overhaul of TDBFPs:

I. Ensure TDBFP ACV and MCV governing characteristics are checked and protocol is signed.
II. Ensure TDBFP P&I checks are done.
III. Ensure all the permits on the system are cancelled.
IV. Ensure that turbine is in decoupled condition from pump before keeping it on barring.
V. Keep LOP & JOP in service & after ensuring healthiness of lube oil circuit, AND partially open the barring gear valve (approx. 20%). Since turbine is in decoupled condition, full opening of barring gear valve will lead to increase of speed of turbine.
VI. Ensure condenser vacuum is pulled up & after charging seal steam connect exhaust valve to the condenser.
VII. Ensure turbine casing drain valves (including man valves) are in open condition.
VIII. Ensure all motorized drain valves in TDBFP extraction steam line are in open condition (including manual valves).
IX. **Following protections are to be bypassed before rolling the turbine:**
    i. TDBFP on & suction flow <100 TPH.
    ii. TDBFP working and suction flow < 250 TPH and R/C valve closed.
    iii. TDBFP tripping from variable set point.
    iv. Suction to discharge DT high.
    v. Speed reference limit to be raised to 6200 rpm
    vi. Discharge valve open feedback to be simulated as it will not allow speed rising beyond 3000rpm.
    vii. Check Speed limit for electrical protection is set to 6300 rpm. (Not to be bypassed under any circumstances)

All other protections (except mentioned above) should be in service**.**

X. Charge control oil circuit & check for any leakages.
XI. Ensure supply is healthy to RTS-1&2 and other valves in the governing rack.
XII. After resetting the turbine, tripping from remote& local push button to be checked & trip oil draining & ESV closure to be ensured.
XIII. Electrical over-speed tripping to be checked before rolling the turbine by simulation.

Responsibility:UCE/HOD(C&I)

XIV. Open aux steam motorized/manual valve by 10% only.
XV. Keep ACV opening fixed at 20%.
XVI. Reset the TDBFP & Roll the turbine up to 1500 rpm. Ensure TSI parameters are within limits. Make a note of the expansions the expansions and casing metal temperature. Overall expansion will be around 2.0mm
XVII. Ensure all speed transmitters are working properly and if any problem is encountered attend the same before moving further.
XVIII. As the casing temp improves raise the speed up to 5500 RPM.
XIX. At 5500 rpm, do the oil injection test. Repeat the test at least 2 to 3 times.
XX. If oil injection test is successful, then raise the speed up to 5800 rpm.
XXI. After reaching 5800 rpm increase the rate of change of speed little further.
XXII. Mechanical over speed device is likely to operate between 5850-6100 rpm.
XXIII. If over speed device doesn't operate even up to 6200 rpm, don’t increase the speed further.
XXIV. Trip the turbine and ask TMD to clean over speed trip device along with oil path. **Responsibility: HOD(TMD)**
XXV. After completion of the test, normalize all the bypassed protections.

Responsibility: UCE / HOD(C&I)

Procedure For Oil Injection Test for TDBFP:

I. Press the lever on the over speed tester device and keep it pressed only.
II. Then rotate the Black Hand wheel on the lever till its extreme position.
III. Then open the manual valve in the control oil line to the directional valve.
IV. Then slowly open the needle valve in the outlet of the directional valve by seeing the tripping device knob on the bearing pedestal. As you open the needle valve at one point the over speed bolt will act and the tripping device will act, and the tripping device knob comes to the lower position. This ensures that the over speed bolt has acted.
V. Now close the needle valve and then the lever valve in the control fluid line.
VI. Then close the black hand wheel in the opposite direction till it comes to the extreme position
VII. Then slowly release the lever which was in depressed condition till now

by observing the trip oil pressure. If the trip oil pressure is dropping, then again depress and it should continue till the trip device on the bearing is in reset position i.e. till it lifts up.

VIII. Once the trip device resets then the lever can be released fully.

8.0 Instructions for Electrical/Electronic System for TDBFP Drive Turbines

Preconditions:

Electronic over speed test of main turbine shall be done once in a year at TDBFP drive turbine speed range 4500RPM (+/- 3%) in turbine warm conditions while taking care of expansions. Test shall be done only after the TDBFP is loaded 50% or more, for at least 3 to 4 hours. Machine should be warmed up sufficiently so that casing expansion are at least 80% of full load expansion /or as per OEM recommendation. Expansion readings are to be taken and logged. This avoids over speeding in cold condition and stress on cold rotor.

Procedure For Electronic Overspeed:
Actual Over speed electrical Test is to be carried out by actually raising the speed of the TDBFP drive turbine to 4500RPM (+/- 3%) and by lowering the Over speed protection limit set point from 6300 RPM to 4500 RPM (+/- 3%). Note down the speed at which the electric over speed trip signal is generated. Later, reset the electrical trip valve to its original trip value (6300 rpm).

If for any reason, actual overspeed test is not carried out once in 12 months, approval of ED(OS) shall be taken for continuing the unit in operation.

Responsibility: HOD (TMD) & HOD (Oprn)

9.0 Responsibility

HOD (Oprn) is responsible for implementation of this LMI.

10.0 Review

The HOP will be responsible for reviewing this LMI after 2 years or as necessary.

11.0 Checklist for Testing to avoid Excessive Overspeed of Prime Movers

Sl.No.	Operations	Frequency of Check/Action Plan	Responsibility	Remarks (Yes/No)
A.	**Checks before conductance of overspeed test of turbine and drive turbine (as applicable)**			
1.	Confirm from O&M Manual for maximum speed at which prime mover must be stopped.	**Record**	Operation	
2	Check for passing of any stop valves	**Before every unit start- up**	Operation	
3	Ensure overhauling of CRH NRV & Extraction NRV's is being carried out and check operation of FC NRV's and CRH NRV. Check for any passing of NRV's.	**During OH and every unit start up**	TMD & Operation	
4	Checking of operation of stop valves and control valves including governing characteristics	**Before every cold start- up**	Operation/TMD	
5	Testing of regulation of hydraulic speed governor after rolling to 3000rpm as per OEM recommendation (7% in case of KWU Machines)	**After OH whenever any work has been carried out in Hydraulic speed Governor**	Operation/TMD	
6	Ensure field breaker is open and machine in de-excited condition	**Before every test**	Operation	
7	When machine is de-synchronised for carrying out over speed test, the cause of any failures to establish a motoring condition shall be investigated and rectified	**Before every test**	Operation & EMD	
8	Ensure casings expansions are more than 80% of full load expansions	**Before every test**	Operation	
9	Compare the rotor speed DCS values with stroboscope readings	**After Overhaul/ before test**	Operation	
10	Ensure emergency trip is manned who can see the achieved speed	**Before every test**	Operation	
11	Carry out Oil injection test	**Monthly & before every overspeed test**	Operation	
B.	**Overspeed test for mechanical/hydraulic/electro- hydraulic governing system of Turbine**			
1	Actual overspeed shall be carried out after every capital overhauling of Main turbine to evaluate turbo-visory parameter condition up to over speed protection limit	**After every Capital overhauling**	Operation	

Sl.No.	Operations	Frequency of Check/Action Plan	Responsibility	Remarks (Yes/No)
2	If any servicing /repair/replacement is carried out in emergency governor or Gap between and striker & lever adjusted during TG OH, then carry out actual over speed test (Prior to Electrical simulated over speed test)	Every time after servicing/repair/ replacement of emergency Governor or gap adjustment between Striker & lever	Operation	
3	During actual overspeed test 1. Speed at which overspeed bolt operated is within range as per O&M manual 2. If speed is not within overspeed range, then action plan to be prepared. 3. If overspeed bolt is not operated/ any problem with overspeed testing unit, then reason for the same to be investigated and rectified before bringing back the unit.	Record speed at which over speed bolt operated Action plan to be prepared. Rectification of the defect	Operation TMD TMD	(Yes/No)
4	Checking of Stand-by system of automatic starting and run up system (as distinct from automatic run up only and even if it is in the stand-by mode)	Once in a Quarter or as recommended by OEM	Operation & C&I	
5	If No servicing/repair/replacement is carried out after actual overspeed test, electrical simulated overspeed test to be carried out at 3000+/-3% rpm	Once in a year	Operation & C&I	
C. Overspeed test for mechanical/hydraulic/electro- hydraulic governing system of Drive Turbine				
1.	Ensure Drive turbine is in decoupled condition	Before every test	Operation & TMD	
2.	Oil injection test to be carried out before over speed test	Once in Two year	Operation	
3.	Overspeed testing is being carried out in interval of Two years	Once in Two year	Operation	
D. Overspeed test of Turbines and Drive Turbines having Electronic governing system				
1.	Ensure OEM recommended procedure and periodicity is being followed	As per OEM recommendations	Operation	
2.	In absence of OEM guidelines, Simulation overspeed test for main turbine to be carried out yearly	Once in a year	Operation	
3.	In absence of OEM guidelines, actual overspeed test for TDBFP drive turbine to be carried out after every capital OH of drive turbine	After every COH	Operation	
4.	In absence of OEM guidelines, simulated overspeed test for TDBFP drive turbine to be carried out once in a year	Once in a year	Operation	

12.0 Annexures

I. Protocol Main TG actual over speed test
II. Protocol for simulation over speed test of Main turbine/TDBFP drive turbine
III. Protocol for oil injection test/Actual over speed of TDBFP drive turbine
IV. Protocol for Main turbine ATT safety device / oil injection
V. Protocol for steam tightness test of turbine stop/control valves of Main turbine.

PROTOCOL FOR MAIN TURBINE ACTUAL OVERSPEED TEST

UNIT: DATE:

Unit was running with______MW load with mills____________in service. Load has been gradually reduced keeping mills_________in service with oil support in__________elevations.MS pressure wasmaintained ksc.

TEST PROCEDURE:

Primary oil pressure: HP/IP secondary oil pressure:

S.NO	ACTIVITY	REMARKS
1	M/C ensured on EHC	
2	ATT for trip devices have been carried out.	
3	Oil Injection Test carried out. Both bolts operated	Primary oil: Test oil:
4	Noted down all important parameters (vibrations / temperatures)	
5	Bus changed over from UT to ST. (For stage-2 only)	
6	Reduce load up to 50-60 MW	
7	Reduce MS pressure up to 55-60 ksc keeping HP/LP Bypass open. HPBP pressure setpoint to be kept in auto otherwise manual setpoint to be reduced near to MS pressure.	
8	Open generator Breaker and De-synchronize the machine. Record time	De-synch:
9	Generator Field Breaker was hand tripped from UCB and Excitation was off.	
10	Turbine speed was maintained at 3000rpm	
11	Recorded all turbovisory parameters.	
12	Raise EHC output to maximum and record maximum TG speed.	
13	Recorded all turbovisory parameters.	
14	Changed over the machine to Hydraulic and EHC was isolated.	
15	Raised the speed governor to allow maximum speed	Max speed:
16	Raised the Turbine speed further by operating the speeder gear lever at governing rack.	
17	Record Time and speed of Machine tripping at over speed.	Speed: Time:
18	Recorded all turbovisory parameters/ alarms.	
19	Ensure TG comes on barring speed and note down coasting down time	

DEPARTMENT	OPERATION	TMD	EMD	C&I	MTP
SIGNATURE					
NAME					
DESIGNATION					

PROTOCOL FOR SIMULATION OVER SPEED TEST OF MAIN TURBINE/TDBFP

UNIT- DATE: -

S.NO.	OBSERVATION	VALUE
1	TURBINE SPEED	
2	SIMULATED SPEED	
3	TURBINE TRIPPED (YES/NO)	
4	ALARM ---OK/NOT OK	
6	OVERSPEED SET POINT AGAIN NORMALISED	

REMARK:

SIGN.			
NAME			
DESIGNATION			
DEPTT	OPERATION	C&I	TMD

PROTOCOL FOR OIL INJECTION TEST/ACTUAL OVER SPEED OF TDBFP DRIVE TURBINE

UNIT- **DATE: -**

OIL INJECTION TEST OF TDBFP TURBINE

S.NO.	OBSERVATION	VALUE	REMARK
1	UNIT LOAD		
2	TDBFP SPEED		
3	TRIP OIL PRESSURE		
4	OVERSPEED BOLT OPERATED(RPM)		
5	TEST OIL PRESSURE		

ACTUAL OVER SPEED OF TDBFP TURBINE

S.NO.	OBSERVATION	VALUE	REMARK
1	OVERSPEED BOLT OPERATED(RPM)		

REMARK:

SIGN.				
NAME				
DESIGNATION				
DEPTT	OPERATION	TMD	C&I	MTP

PROTOCOL FOR MAIN TURBINE ATT SAFETY DEVICE / OIL INJECTION

Unit: Date:

S. N	Checklist	Status
1	Ensure presence of C&I and TMD personnel while doing ATT of safety device/oil injection	
2	Ensure turbine speed near rated speed to get correct test oil pressure.	
3	After ATT of over speed device, ensure test oil pressure zero.	

S. N	**OBSERVATION**	**VALUE**		**REMARK**
1	UNIT LOAD (MW)			
2	TURBINE SPEED (RPM)			
3	PRIMARY OIL PR(KSC)			
4	AUX TRIP OIL PR(KSC)			
5	TRIP OIL PR (KSC)			
6	ATT RTS1 DONE	Y / N		OK / NOT OK
7	ATT RTS2 DONE	Y / N		OK / NOT OK
8	ATT LOW VACUUM DEVICE DONE	Y / N		
	OIL INJECTION TEST DONE FROM			ATT / LOCAL
9	OS BOLT-1 TEST OIL PR		(KSC)	OK / NOT OK ALARM: OK / NOT OK
10	OS BOLT-2 TEST OIL PR		(KSC)	OK / NOT OK ALARM: OK / NOT OK APPEARED IN SOE: Y / N

REMARK (if any):			
SIGN:			
NAME:			
DESIGNATION:			
DEPARTMENT:	OPERATION	TMD	C&I

PROTOCOL STEAM TIGHTNESS TEST OF TURBINE STOP/CONTROL VALVES OF MAINTURBINE

UNIT- DATE: -

SL NO	PARAMETER	VALUE
1	START UP TYPE	
2	MS PRESSURE	
3	MS TEMPERATURE	
4	HRH PRESSURE	
5	HRH TEMPERATURE	
6	INTIAL TURBINE SPEED	
7	FINAL TURBINE SPEED (AFTER OPENING STOP/CONTROL VALVE)	

REMARK:

SIGN.		
NAME		
DESIGNATION		
DEPTT	OPERATION	TMD